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Abstract

The purpose of this project is to develop a process that predicts wind behavior and uses those predictions to control the direction a wind turbine faces. Additional benefits of prediction is the use of speed to determine when a turbine may be operating outside of its optimal range.

Discoveries

Need two nets with two different windows

Only need to turn the turbine 180 degrees

Should keep track of window for which the speed is outside of range

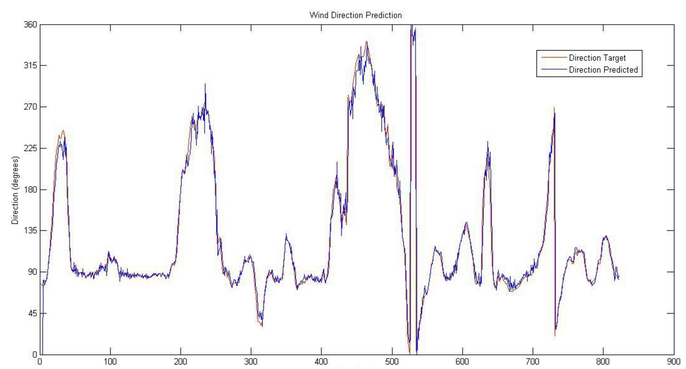
Understand how much electricity required to move turbine

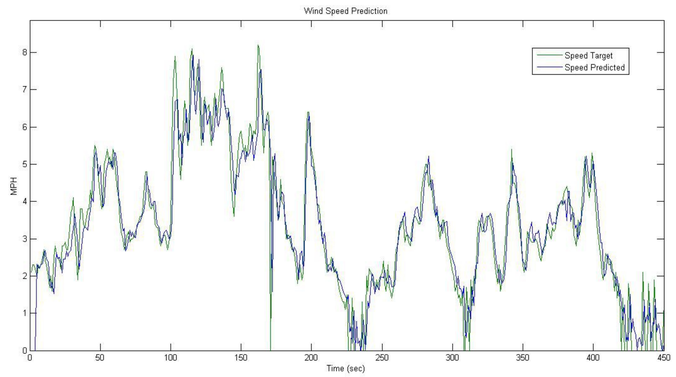
(flow chart)

Build it

Problem

Most traditional horizontal axis wind turbines stay facing one direction, losing maximum angle of attack,. Some turbines do adjust their direction after the wind changes, losing potential energy for electricity generation and increasing wear and tear on the turbine components.





Future Work

My goal is to integrate all aspects of this process (input, neural network analysis, and control signal output) onto a microcontroller running a real time operating system. Eventually it would be necessary to incorporate a networked aspect to accommodate entire wind farms.

Current Methods

WindWorks anemometer and e-vane kits

IMSE software

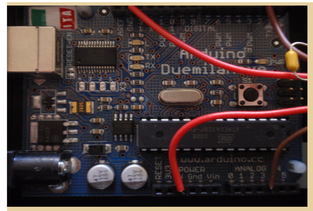
MATLAB's Simulink

MicroC/OS

8086 control board

Approaches

Running collected data through a Self Organizing Map (most accurately predicted changes in wind speed 3 seconds into the future. Changes in wind speed were predicted in a 25 minute window. When predicted output deviates from current wind direction, signal is sent to the board which controls the motor.



(pic)

8086 board

Dry creek turbine